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Assembly table

Abstract

An assembly table includes a table board, a fixing frame and a plurality of detachable table legs. The table board includes an upper surface and a lower surface, and the fixing frame is fixed on the lower surface of the table board. In addition, the fixing frame includes a plurality of connecting devices, connecting rods and connecting rod stability adjustment modules. The connection devices include a plurality of lock protrusions and a plurality of adjusting seats, and a plurality of openings are formed on the connecting rods for locking to the lock protrusions of the connection devices. The connecting rod stability adjustment modules are installed on the connecting devices, and each of the connecting rod stability adjusting modules includes an adjusting screw, pre-locked in a corresponding adjustment seat of the adjustment seats of the connecting devices to stably connect the connecting rods.

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Background/Summary

RELATED APPLICATIONS

(1) This application claims priorities to Taiwan Application Serial Number 111139740, filed Oct. 20, 2022 and Taiwan Application Serial Number 112109404, filed Mar. 14, 2023, the disclosures of which are incorporated herein by reference in their entireties.

TECHNICAL FIELD

(2) The present disclosure relates to an assembly table. More particularly, the present disclosure relates to an assembly table with detachable table legs.

BACKGROUND

(3) With the progress of the society, the designs of various types of furniture are also more diversified. However, due to the dense population of the city, people's living space is gradually narrow. How to make full use of the limited space through proper furniture configuration, release the existing space to store the furniture and use the furniture space in time when needed may provide users with effective space utilization.

(4) Some assembly table can be conveniently stored, and the assembly table can be detached and stored when not in use to reduce the occupied space, and therefore the assembly table has become one of the commonly used equipment at home and in the workplace. In addition, the simple appearance of the assembly table can adapt to various home styles or be used as a work table, a conference table, or even a display table. Therefore, the assembly table can be widely used in home life and in the office.

(5) Therefore, there is a need to provide a more stable and safe assembly table able to flexibly adapt to different size table boards and be conveniently assembled.

SUMMARY

(6) The summary of the present invention is intended to provide a simplified description of the disclosure to enable readers to have a basic understanding of the disclosure. The summary of the present invention is not a complete overview of the disclosure, and it is not intended to point out the importance of the embodiments/key elements of the present invention or define the scope of the invention.

(7) One objective of the embodiments of the present invention is to provide an assembly table able to be conveniently stored, transported, used and accommodated.

(8) To achieve these and other advantages and in accordance with the objective of the embodiments of the present invention, as the embodiment broadly describes herein, the embodiments of the present invention provides an assembly table including a table board, a fixing frame and a plurality of detachable table legs. The table board includes an upper surface and a lower surface, and the fixing frame is fixed on the lower surface of the table board. In addition, the fixing frame includes a plurality of connecting devices, a plurality of connecting rods and a plurality of connecting rod stability adjustment modules. The connecting devices include a plurality of lock protrusions and a plurality of adjusting seats, the connecting rods are connected between the connecting devices, and the connecting rods include a plurality of openings to lock to the lock protrusions of the connecting devices. The connecting rod stability adjustment modules are installed on the connecting devices, and each of the connecting rod stability adjustment modules includes an adjusting screw pre-locked in a corresponding adjustment seat of the adjustment seats of the connecting devices. In addition, a protruding length of the adjusting screw protruding from the corresponding adjusting seat is adjusted to stably connect the connecting rods on the connecting devices after the connecting rods are connected to the connecting devices. In addition, the detachable table legs are detachably installed on the fixing frame.

(9) In some embodiments, the fixing frame further includes a plurality of assembly connecting

modules to fix to another assembly table.

(10) In some embodiments, the assembly connecting modules include a plurality of fixing shafts and a plurality of rotary lock devices. The fixing shafts are fixed on the connecting devices and the rotary lock devices are pivoted on some of the connecting devices to lock fixing shafts of the another assembly table.

(11) In some embodiments, each of the rotary lock devices includes a rotating ring, an extension rod, a fastening hook and an operating handle. The rotating ring is pivoted on one corresponding fixing shaft of the connecting devices, the extension rod is connected to the rotating ring, the fastening hook is connected to the extension rod to lock a fixing shaft on the another assembly table and the operating handle is connected to the fastening hook to rotate the fastening hook.

(12) In some embodiments, the connecting devices are angle adjustable connecting devices fixed on the lower surface of the table board and angles the angle adjustable connecting devices are adjusted according to a shape of the table board and lengths of the connecting rods.

(13) In some embodiments, the connecting rods include at least one first connecting rod and at least one second connecting rod. The lengths of the first connecting rod and the second connecting rod are different.

(14) In some embodiments, each of the angle adjustable connecting device includes a first angle adjustment unit and a second angle adjustment unit. The first angle adjustment unit is connected to the first connecting rod and the second angle adjustment unit is connected to the second connecting rod and engaged with the first angle adjustment unit to determine and fix an angle between the first connecting rod and the second connecting rod.

(15) In some embodiments, the first angle adjustment unit includes a first rotating part, a connecting rod fixing part and a first engaging gear. The connecting rod fixing part is connected to the first rotating part and locks the first connecting rod. The first engaging gear is formed on the first rotating part. In addition, the second angle adjustment unit includes a second rotating part, a connecting rod fixing part and a second engaging gear. The connecting rod fixing part is connected to the second rotating part and locks the second connecting rod and the second engaging gear is formed on the second rotating part to engage with the first engaging gear so as to fix the angle between the first connecting rod and the second connecting rod.

(16) In some embodiments, each of the detachable table legs includes a table leg main body and a fixing device. The fixing device is connected to one end of the table leg main body to lock in a fixing hole of the first rotating part to clamp the second rotating part between the table leg main body and the first rotating part.

(17) In some embodiments, the fixing device is a quick release fixing device to engage in a fixing slot of the first rotating part so as to clamp the second rotating part between the table leg main body and the first rotating part.

(18) Hence, the assembly table according to some embodiments of the present invention can not only be stored and accommodated conveniently, but also can be installed and moved conveniently, and the angle adjustment unit is locked by the detachable table legs to tightly fix the angle adjustment unit on the fixing frame, and the strength and stability of the assembly table may be further increased by the connecting rod stability adjustment module. In addition, the fixing frame can be pre-fixed on the bottom surface of the table board, and the fixing frame does not need to be removed from the bottom surface of the table board when disassembling the table legs so that the assembly table may be more conveniently assembled and stored.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The disclosure can be more fully understood by reading the following detailed description of

the embodiment, with reference made to the accompanying drawings as follows:

(2) FIG. 1 illustrates an exploded diagram showing an assembly table according to one embodiment of the present invention;

(3) FIG. 2 illustrates a schematic perspective diagram showing a lower surface of the assembly table;

(4) FIG. 3 illustrates an exploded diagram showing a connecting device of the assembly table;

(5) FIG. 4 illustrates a schematic diagram showing a plurality of assembly tables connected together;

(6) FIG. 5 illustrates a schematic diagram showing a plurality of assembly tables stacked together;

(7) FIG. 6 illustrates a schematic diagram showing a quick release fixing device of the assembly table; and

(8) FIG. 7 illustrates a schematic diagram showing a connecting rod stability adjustment module of the assembly table.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

(9) The following is a detailed description of the embodiments in conjunction with the accompanying drawings, but the provided embodiments are not intended to limit the scope of the disclosure, and the description of the structure and operation is not used to limit the execution sequence thereof. The structure of the recombination of components and the resulting devices with equal functions are all within the scope of this disclosure. In addition, the drawings are for illustration purposes only, and are not drawn according to the original scale. For ease of understanding, the same reference numbers are used in the drawings and the description to refer to the same or like parts.

(10) In addition, the terms used in the entire description and the scope of the patent application, unless otherwise specified, usually have the usual meaning of each term used in this field, in the content disclosed here and in the special content. Some terms used to describe the disclosure are discussed below or elsewhere in this specification to provide additional guidance to those skilled in the art in the disclosure.

(11) In the implementation mode and the scope of the present application, unless the article is specifically limited in the context, “a” and “the” can generally refer to a single or pluralities. In the steps, the numbering is only used to conveniently describe the steps, rather than to limit the sequence and implementation.

(12) Secondly, the words “comprising”, “including”, “having”, “containing” and the like used in the present application are all open language, meaning including but not limited to.

(13) FIG. 1 illustrates an exploded diagram of the assembly table according to one embodiment of the present invention, FIG. 2 illustrates a schematic perspective diagram of the lower surface of the assembly table, FIG. 3 illustrates an exploded diagram of the connecting device of the assembly table, FIG. 4 illustrates a schematic diagram of a plurality of assembly tables connected together, FIG. 5 illustrates a schematic diagram of a plurality of assembly tables stacked together, and FIG. 6 illustrates a schematic diagram of the quick release fixing device of the assembly table. In addition, FIG. 7 illustrates a schematic diagram of the connecting rod stability adjustment module of the assembly table.

(14) First referring to FIG. 1 and FIG. 2, the assembly table **100** includes a table board **110**, a fixing frame **120** and a plurality of detachable table legs **130**. The table board **110** includes an upper surface **112** and a lower surface **114**, and the fixing frame **120** is fixed on the lower surface **114** of the table board **110**. In addition, the fixing frame **120** includes a plurality of connecting devices **125** and a plurality of connecting rods, for example, a first connecting rod **122** and a second connecting rod **124**. The connecting rods are connected between the connecting devices **125**.

(15) In some embodiments, the connecting devices **125** is an angle adjustable connecting device **126** able to adjust the angle thereof according to the shape of the table board **110** and the lengths of the connecting rods so as to conveniently fix on the lower surface **114** of the table board **110**.

(16) In some embodiments, three connecting rods and three angle adjustable connecting devices **126** may be form a triangle fixing frame. In some embodiments, four connecting rods and four angle adjustable connecting devices **126** may be formed a rectangular fixing frame or any kind of quadrilateral fixing frame without departing from the spirit and scope of the disclosure. The fixing frame **120** may be formed into any shape according to the actual shape of the table board **110** with required connecting rods and angle adjustable connecting device **126** to support the table board **110**, and users may adjust the shape of the fixing frame **120** according to the actual demand without departing from the spirit and scope of the disclosure.

(17) In some embodiments, the lengths of the first connecting rod **122** and the second connecting rod **124** are different.

(18) In addition, a plurality of detachable table legs **130** are detachably installed on the fixing frame **120**, and fixed on the angle adjustable connecting device **126** of the fixing frame **120** to effectively support the table board **110** and the fixing frame **120**.

(19) In some embodiments, the connecting device **125** may be a fixed connecting device connected between the first connecting rod **122** and the second connecting rod **124** without departing from the spirit and scope of the disclosure.

(20) In some embodiments, the fixing frame **120** further includes a plurality of assembly connecting modules **128** to fix to other assembly tables **100**.

(21) Simultaneously referring to FIG. 4, as shown in the figures, the assembly connecting module **128** includes a plurality of fixing shafts **410** and a plurality of rotary lock devices **420**. The fixing shafts **410** are fixed on the connecting devices **125**, i.e. the angle adjustable connecting devices **126**, and the rotary lock device **420** are pivoted on some angle adjustable connecting devices **126** to lock to corresponding fixing shafts **410** on another assembly table **100**. Three assembly tables **100A**, **100B** and **100C** are illustrated as an example, corners of the first assembly table **100A**, the second assembly table **100B**, and the third assembly table **1000** are shown to further explain the assembly connecting module **128**.

(22) In some embodiments, each of the rotary lock devices **420** includes a rotating ring **422**, an extension rod **424**, a fastening hook **426** and an operating handle **428**. The rotating ring **422** is pivoted on the fixing shaft **410** so as to able to rotate on the fixing shaft **410**. The extension rod **424** is connected to the rotating ring **422**, the fastening hook **426** is connected to the extension rod **424** to lock to the fixing shaft **410** of the other assembly table **100**. Taking the rotary lock device **420** of the first assembly table **100A** as an example, the rotary lock device **420** of the first assembly table **100A** may rotate about 180 degrees to lock to the fixing shaft **410** of the second assembly table **100B**. The rotary lock device **420** of the second assembly table **100B** may also rotate about 180 degrees to lock to the fixing shaft **410** of the third assembly table **1000**. In addition, the rotary lock device **420** of the third assembly table **1000** is be rotated about 165 degrees to expose the rotary lock device **420** outside the lower surface of the third assembly table **100C** to show the appearance shape of the rotary lock device **420**.

(23) In some embodiments, the rotary lock device **420** is preferably stored under the lower surface **114** of the table board **110** rather than being exposed outside the lower surface **114** of the table board **110** of the third assembly table **1000** as illustrated on FIG. 4 in order to prevent users from inadvertently touching the rotary lock device **420** or the rotary lock device **420** hitting the surface of the other object to cause injury to the user or the equipment. In some embodiments, the rotary lock device **420** preferably includes a fixing element to fix and hide the rotary lock device **420** under the lower surface **114** of the table board **110**. When using the assembly table **100**, the user may rotate the rotary lock device **420** to the outside the lower surface **114** of the table board **110** to conveniently lock the rotary lock device **420** to the fixing shaft **410** of another assembly table **100**.

(24) In addition, the operating handle **428** is connected to the fastening hook **426** so that the user may conveniently rotate the rotary lock device **420** to lock the fastening hook **426** to the fixing shaft **410** of another assembly table **100**.

(25) Referring to FIG. 3, as shown in the figure, in some embodiments, the angle adjustable connecting device **126** includes a first angle adjustment unit **310** and a second angle adjustment unit **320**. The first angle adjustment unit **310** is detachably connected to the first connecting rod **122**, the second angle adjustment unit **320** is detachably connected to the second connecting rod **124**, and the first angle adjustment unit **310** and the second angle adjustment unit **320** may adjust and fix the angle between the first connecting rod **122** and the second connecting rod **124**.

(26) In some embodiments, the first angle adjustment unit **310** includes a first rotating part **312**, a connecting rod fixing part **314** and a first engaging gear **316**. The second angle adjustment unit **320** includes a second rotating part **322**, a connecting rod fixing part **324** and a second engaging gear **326**.

(27) The connecting rod fixing part **314** is connected to the first connecting rod **122**. In some embodiments, the connecting rod fixing part **314** is locked to the first connecting rod **122**. Another end of the connecting rod fixing part **314** is connected to the first rotating part **312**, and the first engaging gear **316** is formed on the first rotating part **312**.

(28) In addition, the connecting rod fixing part **324** of the second angle adjustment unit **320** is connected to the second connecting rod **124**. In some embodiments, the connecting rod fixing part **324** of the second angle adjustment unit **320** is locked to the second connecting rod **124**. Another end of the connecting rod fixing part **324** is connected to the second rotating part **322**, and the second engaging gear **326** is formed on the second rotating part **322** and engaged with the first engaging gear **316** to effectively fix the angle between the first connecting rod **122** and the second connecting rod **124**.

(29) In some embodiments, the first engaging gear **316** is an internal gear, and the second engaging gear **326** is an external gear, but not limited thereto. In some embodiments, the first engaging gear **316** may be a groove and the second engaging gear **326** may be a protrusion, or the first engaging gear **316** may be a protrusion and the second engaging gear **326** may be a groove without departing from the spirit and scope of the present invention.

(30) In some embodiments, the detachable table leg **130** includes a table leg main body **132** and a fixing device **134**. The fixing device **134** is connected to one end of the table leg main body **132**, penetrated into the table leg fixing hole **202**, and fixed in the fixing hole **318** of the first rotating part **312** so that the second rotating part **322** is clamped between the table leg main body **132** and the first rotating part **312**.

(31) In some embodiments, the fixing hole **318** has a female thread, and the fixing device **134** of the detachable table legs **130** has a male thread to lock each other so that the first engaging gear **316** of the first rotating part **312** and the second engaging gear **326** of the second rotating part **322** are firmly engaged together, thereby enhancing the rigidity of the fixing frame **120** and improving the strength and stability of the assembly table **100**.

(32) In some embodiments, simultaneously referring to FIG. 1 to FIG. 3, the second connecting rod **124** is formed by a rectangular tube **340**, and a table board fixing hole **201**, a fixing hole **344** and an openings **342** are formed thereon. In addition, the connecting rod fixing part **324** of the second angle adjustment unit **320** includes a lock protrusion **327** and a fixing hole **329**, the lock protrusion **327** may lock to the opening **342** of the rectangular tube **340** to fix the second connecting rod **124** on the connecting rod fixing part **324** of the second angle adjustment unit **320**. At this moment, the fixing hole **329** of the connecting rod fixing part **324** is aligned with the fixing hole **344** and the table board fixing hole **201** of the rectangular tube **340**.

(33) Because the lock protrusion **327** of the connecting rod fixing part **324** is locked to the opening **342** of the rectangular tube **340**, the second connecting rod **124** may be conveniently installed on the second angle adjustment unit **320** or removed from the second angle adjustment unit **320** so as to further improve the convenience of assembling the assembly table **100**.

(34) In addition, the table board fixing hole **201** is preferably larger than the fixing hole **344**, and the fixing hole **329** is a countersunk hole, the lower hole diameter is larger than the upper hole

diameter, so that a fixing device, such as a screw, may pass through the table board fixing hole **201** and be fixed in the fixing hole **329** of the connecting rod fixing part **324**, and therefore the second connecting rod **124** and the second angle adjustment unit **320** are effectively fixed on the lower surface **114** of the table board **110**.

(35) In the same manner, the first connecting rod **122** is formed by a rectangular tube **330**, and a table board fixing hole **201**, a fixing hole **334** and an openings **332** are formed thereon. In addition, the connecting rod fixing part **314** of the first angle adjustment unit **310** includes a lock protrusion **317** and a fixing hole **319**, the lock protrusion **317** may lock to the opening **332** of the rectangular tube **330** to fix the first connecting rod **122** on the connecting rod fixing part **314** of the first angle adjustment unit **310**. At this moment, the fixing hole **319** of the connecting rod fixing part **314** is aligned with the fixing hole **334** and the table board fixing hole **201** of the rectangular tube **330**.

(36) Because the lock protrusion **317** of the connecting rod fixing part **314** is locked to the opening **332** of the rectangular tube **330**, the first connecting rod **122** may be conveniently installed on the first angle adjustment unit **310** or removed from the first angle adjustment unit **310** so as to further improve the convenience of assembling the assembly table **100**.

(37) In addition, the table board fixing hole **201** is preferably larger than the fixing hole **334**, and the fixing hole **319** is a countersunk hole, the lower hole diameter is larger than the upper hole diameter, so that a fixing device, such as a screw, may pass through the table board fixing hole **201** and be fixed in the fixing hole **319** of the connecting rod fixing part **314**, and therefore the first connecting rod **122** and the first angle adjustment unit **310** are effectively fixed on the lower surface **114** of the table board **110**.

(38) In addition, after the first connecting rod **122** is engaged with the first angle adjustment unit **310** and the second connecting rod **124** is engaged with the second angle adjustment unit **320**, a user may conveniently engage the first engaging gear **316** of the first rotating part **312** of the first angle adjustment unit **310** with the second engaging gear **326** of the second rotating part **322** of the second angle adjustment unit **320** according to a required angle, and fix the first connecting rod **122**, the first angle adjustment unit **310**, the second connecting rod **124** and the second angle adjustment unit **320** on the lower surface **114** of the table board **110** with the fixing devices such as the screws.

(39) The user may screw the fixing device **134** of the detachable table legs **130** into the fixing hole **318** of the first rotating part **312** along the table leg fixing hole **202** through the through hole **328** of the second rotating part **322** so that the second rotating part **322** are clamped between the table leg main body **132** and the first rotating part **312** to enhance the strength and stability of the assembly table **100**.

(40) In some embodiments, the detachable table leg **130** further includes a sliding wheel **136** installed on another end of the table leg main body **132** to conveniently move the assembly table **100**. However, the present invention is not limited thereto, the another end of the table leg main body **132** may also be equipped with a height adjuster or an anti-slip pad, without departing from the spirit and scope of the present invention.

(41) In some embodiments, the assembly table **100** further includes a table leg storage device **160** fixed on the lower surface **114** of the table board **110** to accommodate the detachable table legs **130** while the assembly table **100** is stored. The table leg storage device **160** includes a fixing base **162** and a plurality of hook claws **164**. The fixing base **162** is fixed on the lower surface **114** of the table board **110**, and the hook claws **164** are formed on the fixing base **162** to accommodate and fix the detachable table legs **130**. In some embodiments, two table leg storage devices **160** are preferably fixed on two ends of the lower surface **114** of the table board **110** to stably fix the detachable table legs **130**.

(42) Referring to FIG. 5, a first assembly table **100A** and a second assembly table **100B** are stored and stacked together as an example. In some embodiments, the fixing shafts **410** of the first assembly table **100A** may be stacked on the upper surface **112** of the table board **110** of the second

assembly table **1001B**, and the fixing shafts **410** of the second assembly table **1001B** may be placed on the ground or on an upper surface of another assembly table to effectively protect the surface of the table board **110** from being scratched and facilitate storage or movement. The fixing shaft **410** is protruded on the fixing frame **120** and the detachable table legs **130** are stored in the fixing frame **120** so that the detachable table legs **130** is less than the height of the fixing shaft **410** measured from the fixing frame **120** so that the detachable table legs **130** can be effectively protected by the fixing frame **120** to prevent from the detachable table legs **130** from being damaged during storage or moving, thereby increasing the service life of the assembly table **100**.

(43) In some embodiments, referring to FIG. 6, the detachable table leg **630** includes a table leg main body **632**, a quick release fixing device **634** and a sliding wheel **636**. The quick release fixing device **634** may quickly be fixed in the fixing slot **650** of the first rotating part **612** so that the second rotating part **622** is clamped between the table leg main body **632** and the first rotating part **612** to increase the strength and stability of the fixing frame and conveniently install and remove the detachable table legs **630**.

(44) In some embodiments, referring to FIG. 7, the connecting device **125** further includes a connecting rod stability adjustment module **700** to stably fix the first connecting rod **122** and the second connecting rod **124** to the connecting device **125**. In addition, the connecting device **125** can be an fixed angle connecting device or an angle adjustable connecting device **126**.

(45) The angle adjustable connecting device **126** is illustrated as an example. The first angle adjustment unit **310** and the second angle adjustment unit **320** respectively include a connecting rod stability adjustment module **700** to stably fix the first connecting rod **122** and the second connecting rod **124** on the first angle adjustment unit **310** and the second angle adjustment unit **320**. As shown in the figure, the connecting rod stability adjustment module **700** includes an adjusting screw **710** pre-locked in a corresponding adjustment seat **720**. The table board fixing holes **201** of the first connecting rod **122** and the second connecting rod **124** are respectively aligned with the fixing hole **319** of the first angle adjustment unit **310** and the fixing hole **329** of the second angle adjustment unit **320**, and then the first angle adjustment unit **310** and the second angle adjustment unit **320** are fixed on the lower surface **114** of the table board **110** with the fixing devices such as screws. At this moment, the connecting rod adjusting hole **203** is aligned with the countersunk hole **712**, i.e. a hexagonal countersunk hole, to allow the user using a tool such as a hexagonal wrench to adjust the height of the adjusting screw **710** protruding from the adjusting seat **720** through the connecting rod adjusting hole **203**. The upper surface of the adjusting screw **710** may abut against the inner surfaces of the first connecting rod **122** and the second connecting rod **124** to tightly fit the first connecting rod **122** and the second connecting rod **124** on the first angle adjustment unit **310** and the second angle adjustment unit **320** by the pressure exerted by the adjusting screw **710** so as to prevent from sliding or rotation of the first connecting rod **122** and the second connecting rod **124** on the first angle adjustment unit **310** and the second angle adjustment unit **320**, thereby improving the stability of the first connecting rod **122**, the second connecting rod **124**, the first angle adjustment unit **310** and the second angle adjustment unit **320** and further improving the stability of the assembly table **100**. Therefore, the connecting rod stability adjustment module **700** may stably fix the first connecting rod **122** and the second connecting rod **124** on the connecting device **125**. After the openings of the first connecting rod **122** and the second connecting rod **124** are respectively locked with the lock protrusions of the connecting devices **125**, the connecting rod stability adjustment module **700** may further stably fix the first connecting rod **122** and the second connecting rod **124** on the connecting devices **125** to effectively improve the stability of the first connecting rod **122** and the second connecting rod **124** on the connecting devices **125** so as to further improve the stability of the assembly table **100**.

(46) Accordingly, the assembly table according to some embodiments of the present invention can not only be stored and accommodated conveniently, but also can be installed and moved conveniently, and the angle adjustment unit is locked by the detachable table legs to tightly fix the

angle adjustment unit on the fixing frame, and the strength and stability of the assembly table may be further increased by the connecting rod stability adjustment module. In addition, the fixing frame can be pre-fixed on the bottom surface of the table board, and the fixing frame does not need to be removed from the bottom surface of the table board when disassembling the table legs so that the assembly table may be more conveniently assembled and stored.

(47) Although the present disclosure has been disclosed above in terms of implementation, it is not intended to limit the present disclosure. Any person with ordinary knowledge in the field may make various variations and modifications without departing from the spirit and scope of the disclosure. Therefore, the spirit and scope of the appended claims should not be limited to the description of the embodiments contained herein.

Claims

1. An assembly table, comprising: a table board, comprising an upper surface and a lower surface; a fixing frame fixed on the lower surface of the table board, wherein the fixing frame comprises: a plurality of connecting devices, wherein the connecting devices comprise a plurality of lock protrusions and a plurality of adjusting seats; a plurality of connecting rods connected between the connecting devices, wherein the connecting rods comprise a plurality of connecting rod adjusting holes and a plurality of openings to lock to the lock protrusions of the connecting devices; and a plurality of connecting rod stability adjustment modules installed on the connecting devices, wherein each of the connecting rod stability adjustment modules comprises an adjusting screw pre-locked in a corresponding adjustment seat of the adjustment seats of the connecting devices, wherein a diameter of the adjusting screw is greater than a diameter of a corresponding connecting rod adjusting hole of the connecting rod adjusting holes, wherein a protruding length of the adjusting screw protruding from the corresponding adjusting seat is adjusted to allow the adjusting screw to press on an inner surface of an corresponding connecting rod of the connecting rods so as to stably connect the connecting rods on the connecting devices after the connecting rods are connected to the connecting devices, and the adjusting screw is hidden in the corresponding connecting rod adjusting hole; and a plurality of detachable table legs detachably installed on the fixing frame.
2. The assembly table of claim 1, wherein the fixing frame further comprises a plurality of assembly connecting modules to fix to another assembly table.
3. The assembly table of claim 2, wherein the assembly connecting modules comprise: a plurality of fixing shafts fixed on the connecting devices; and a plurality of rotary lock devices pivoted on some of the connecting devices to lock fixing shafts of the another assembly table.
4. The assembly table of claim 3, wherein each of the rotary lock devices comprises: a rotating ring pivoted on one corresponding fixing shaft of the connecting devices; an extension rod connected to the rotating ring; a fastening hook connected to the extension rod to lock a fixing shaft on the another assembly table; and an operating handle connected to the fastening hook to rotate the fastening hook.
5. The assembly table of claim 1, wherein the connecting devices are angle adjustable connecting devices fixed on the lower surface of the table board and angles the angle adjustable connecting devices are adjusted according to a shape of the table board and lengths of the connecting rods.
6. The assembly table of claim 5, wherein the connecting rods comprise: at least one first connecting rod; and at least one second connecting rod, and lengths of the first connecting rod and the second connecting rod are different.
7. The assembly table of claim 6, wherein each of the angle adjustable connecting device comprises: a first angle adjustment unit connected to the first connecting rod; and a second angle adjustment unit connected to the second connecting rod and engaged with the first angle adjustment unit to determine and fix an angle between the first connecting rod and the second connecting rod.

8. The assembly table of claim 7, wherein the first angle adjustment unit comprises: a first rotating part; a connecting rod fixing part connected to the first rotating part and locking the first connecting rod; and a first engaging gear formed on the first rotating part; wherein the second angle adjustment unit comprises: a second rotating part; a connecting rod fixing part connected to the second rotating part and locking the second connecting rod; and a second engaging gear formed on the second rotating part to engage with the first engaging gear so as to fix the angle between the first connecting rod and the second connecting rod.

9. The assembly table of claim 8, wherein each of the detachable table legs comprises: a table leg main body; and a fixing device connected to one end of the table leg main body to lock in a fixing hole of the first rotating part to clamp the second rotating part between the table leg main body and the first rotating part.

10. The assembly table of claim 9, wherein the fixing device is a quick release fixing device to engage in a fixing slot of the first rotating part so as to clamp the second rotating part between the table leg main body and the first rotating part.
